

AI & ML Learning Roadmap for Node.js Developers

1. Python Basics

- Syntax, Data Types, Loops, Functions
- List Comprehensions, Lambda functions
- OOPs Concepts in Python
- Modules and Virtual Environments

2. Math & Stats

- Linear Algebra (Vectors, Matrices)
- Probability, Bayes Theorem
- Descriptive & Inferential Statistics
- Gradient Descent Intuition

3. Data Preprocessing

- Pandas, NumPy, DataFrames
- Missing Values, Outlier Detection
- Label Encoding, One-Hot Encoding
- Feature Scaling (MinMax, StandardScaler)

4. ML Algorithms

- Linear & Logistic Regression
- KNN, SVM, Decision Trees, Random Forest
- Naive Bayes, XGBoost
- Clustering: K-Means, Hierarchical

5. Model Evaluation

- Confusion Matrix, Precision, Recall, F1
- ROC-AUC, Cross Validation
- Bias-Variance Tradeoff
- Hyperparameter Tuning (GridSearchCV)

6. Deep Learning

- Perceptrons, Activation Functions
- Neural Networks using TensorFlow/Keras

AI & ML Learning Roadmap for Node.js Developers

- CNNs for image classification
- RNNs & LSTMs for sequential data

7. NLP

- Text Preprocessing (Tokenization, Lemmatization)
- Bag of Words, TF-IDF
- Sentiment Analysis
- Transformers & HuggingFace (Intro)

8. Projects

- Spam Classifier
- House Price Predictor
- Sentiment Analysis App
- Deploy ML model using Flask & Docker

9. Deployment & MLOps

- Saving models with Pickle/Joblib
- Building REST API with Flask/FastAPI
- Dockerize ML apps
- Basic CI/CD for ML pipeline